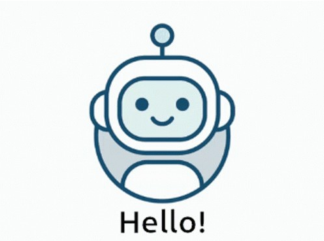

SkyInsight

Predictive Passenger Experience Analytics & Loyalty Modeling

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- Strategic Goals
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- Service Deep Dive
- Loyal but dissatisfied
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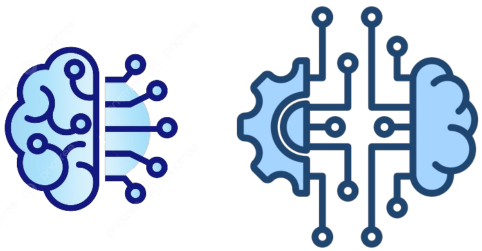
- Model Selection: Random Forest vs XGBoost

Illustrations by Pixeltrue on icons8



Overview

This project delivers a machine learning model to classify passenger satisfaction and predict customer churn. By identifying critical travel pain points, the system provides data-driven insights to improve airline services and boost retention.

Business perspective

- **The Core Challenge:** While 82% of the passenger base consists of loyal customers, 31% within this high-value segment are dissatisfied, posing a significant hidden churn threat.
- **Project Scope:** Analyse data-driven insights to uncover the root causes of loyal customer dissatisfaction and predict churn risk.
- **Strategic Goal:** Deploy targeted interventions to secure and maximize the passenger retention rate.

Resources

01 Dataset

The analysis of 130,000 passenger surveys to understand customer behaviour.

02 Features

It includes 23 features, such as customer demographics, online services, inflight entertainment, cabin comfort and flight delays.

03 ML modelling

The model analyses features and patterns to predict customer satisfaction and potential churn.

04 Business Impact

Long-term customer loyalty: driving cost savings and revenue protection to boost airline profitability.

From Reactive to Proactive Service

Loyalty Stagnation

Traditional survey methods are reactive, feedback arrives too late, after the customer is already dissatisfied.

Rising Acquisition Costs

Acquiring a new passenger is 5-7x more expensive than retaining an existing one.

Data Blind Spots

High-level dissatisfaction is visible, but the critical impact of specific touchpoints across customer segments remains unknown.

Data-Driven Insights

Deep-dive analysis of historical passenger satisfaction data.

Predictive ML Deployment

Predicting passenger satisfaction to identify "at-risk" segments early.

Investment Prioritization

Pinpointing high-impact features to maximize Return on Investment (ROI).

Strategic Goals

Loyalty & retention

Mitigating churn by proactively predicting passenger dissatisfaction.

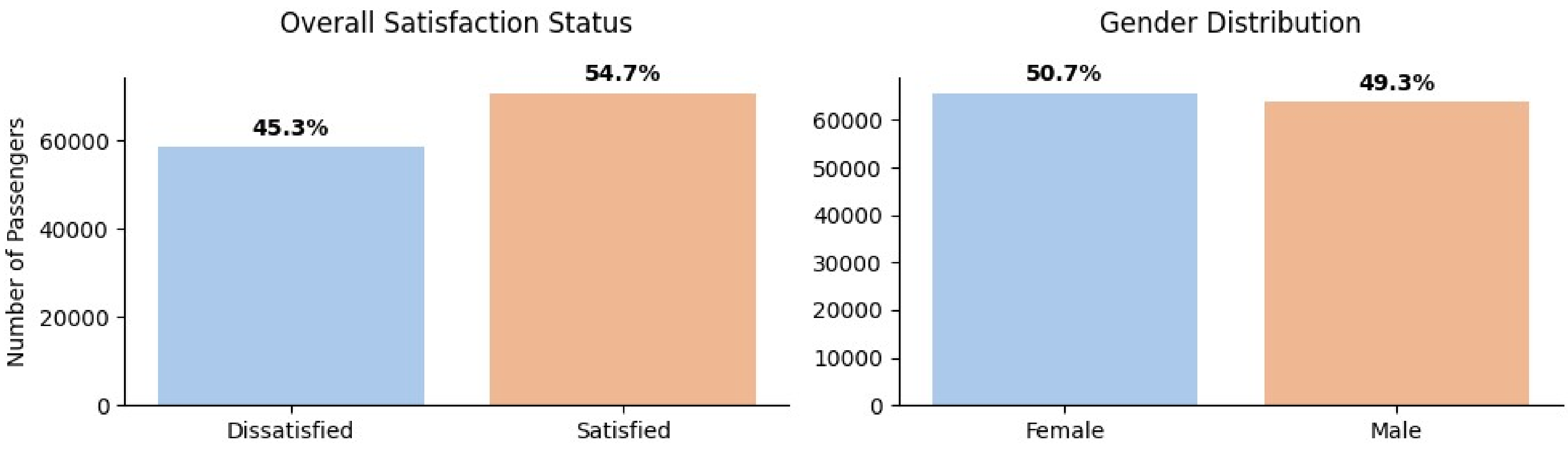
Customer as promoter

Net Promoter Score (NPS): loyal customer who become promoters are the engine of sustainable growth.

Budget Optimization

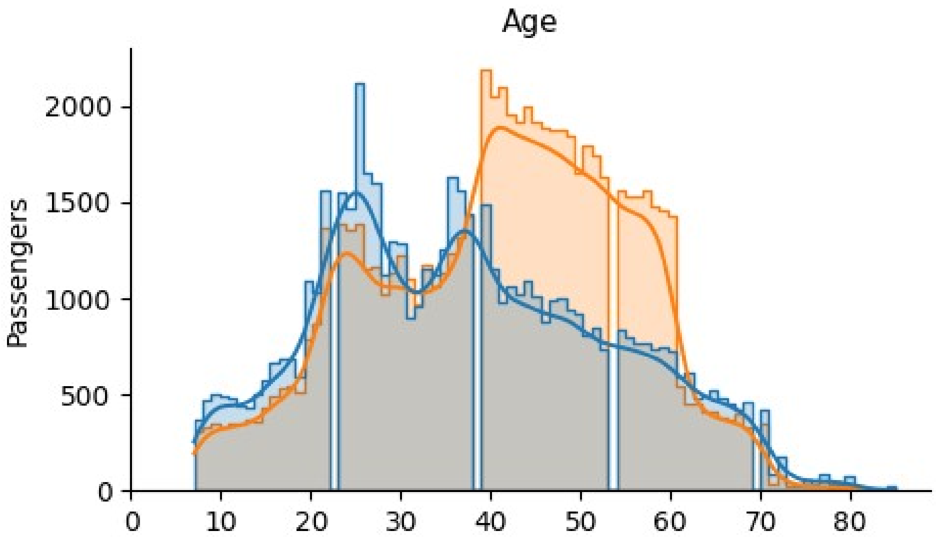
This model safeguards the company’s revenue, reducing the loss of loyal customers and targeting our investment in the right services, rather than wasting the budget.

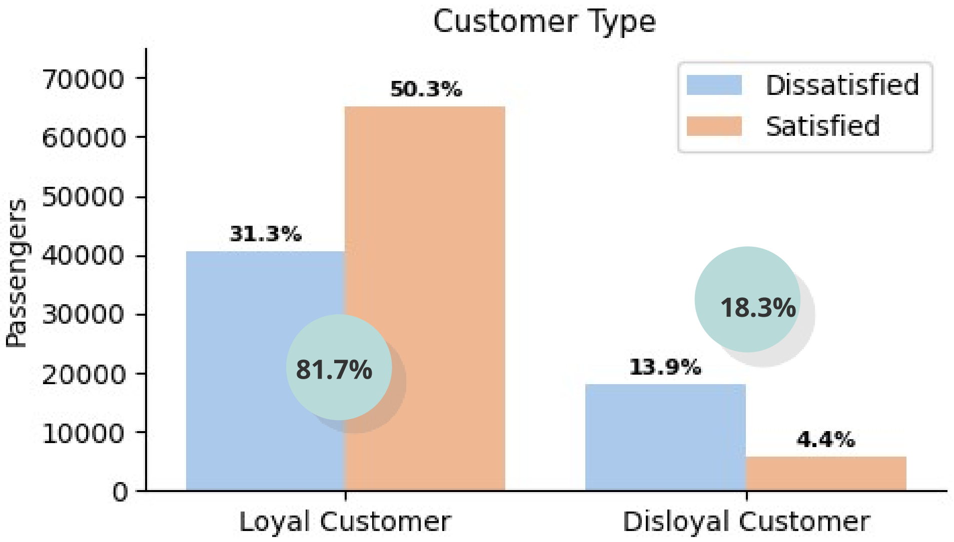
Customer Analysis: Understanding Loyalty Drivers



- **Customer Satisfaction:** 55% Satisfied, 45% at risk of customer churn.
- **Gender Balance:** Equal representation of men and women across all segments and service categories.

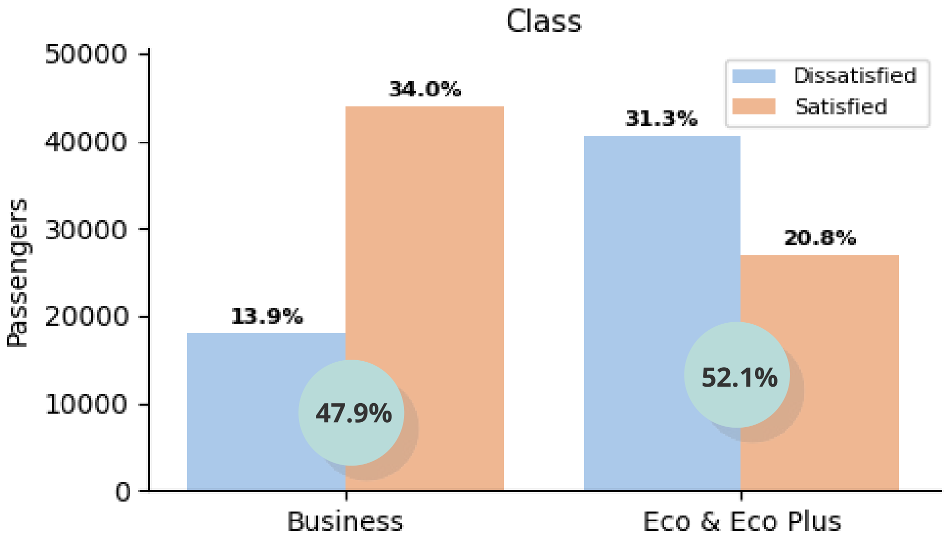
Customer Analysis: Understanding Loyalty Drivers

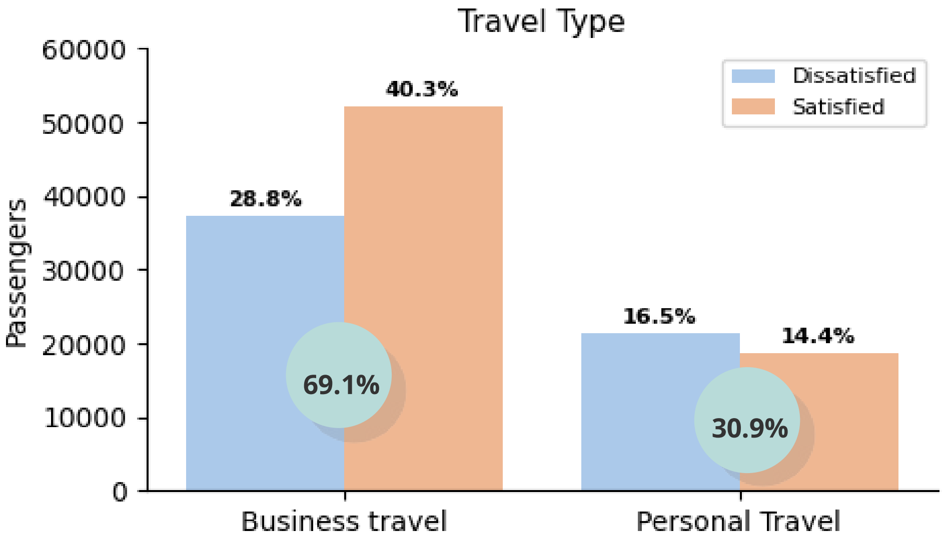




- **Age Satisfaction:** Highest satisfaction observed in the 38–60 age group. Younger ages are dissatisfied.
- **Loyalty Gap:** While 82% of passengers are loyal, nearly 31% remain dissatisfied. Retention of loyalty is the #1 priority.

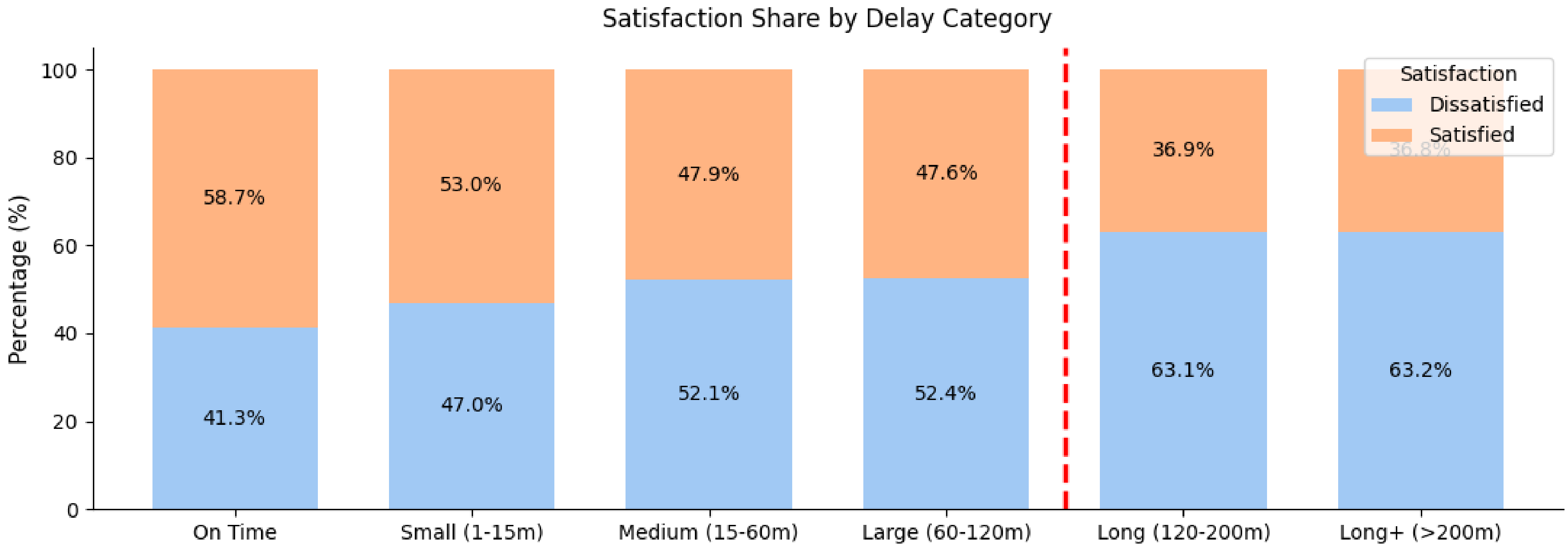
Customer Analysis: Understanding Loyalty Drivers





- **Class Advantage:** In Business class 48% vs 52% in Eco class. Business more satisfied compared to Eco classes.
- **Travel Type:** Business travelers make up 69% of our passengers, yet 29% of them remain dissatisfied. Personal travelers are even less satisfied.
- **Growth Opportunity:** Eco classes and Personal Travel are the primary sources of negative feedback.

Customer Analysis: Understanding Loyalty Drivers



- **Critical Threshold:** 15 minutes. Beyond this mark, dissatisfied passengers become the majority 52%+.
- **Point of No Return:** 120 minutes. After a 2-hour delay, dissatisfaction spikes to 63% and plateaus.
- **Issue:** 56% of flights arrive on time, but 41% of these passengers is dissatisfied. Punctuality alone isn't enough.
- **Business Takeaway:** Focus on preventing delays under 60 min. Once a delay exceeds 2h, loyalty is compromised.

Proactive Retention: From ML Predictions to Business Impact

Predictive Insights

Early Warning Detection:

• ML model identifies high-risk passengers before dissatisfaction leads to churn.

• Algorithms pinpoint exactly who is at risk and why.

Targeted Interventions

Data-Driven Actions:

• Systemic upgrades are deployed.

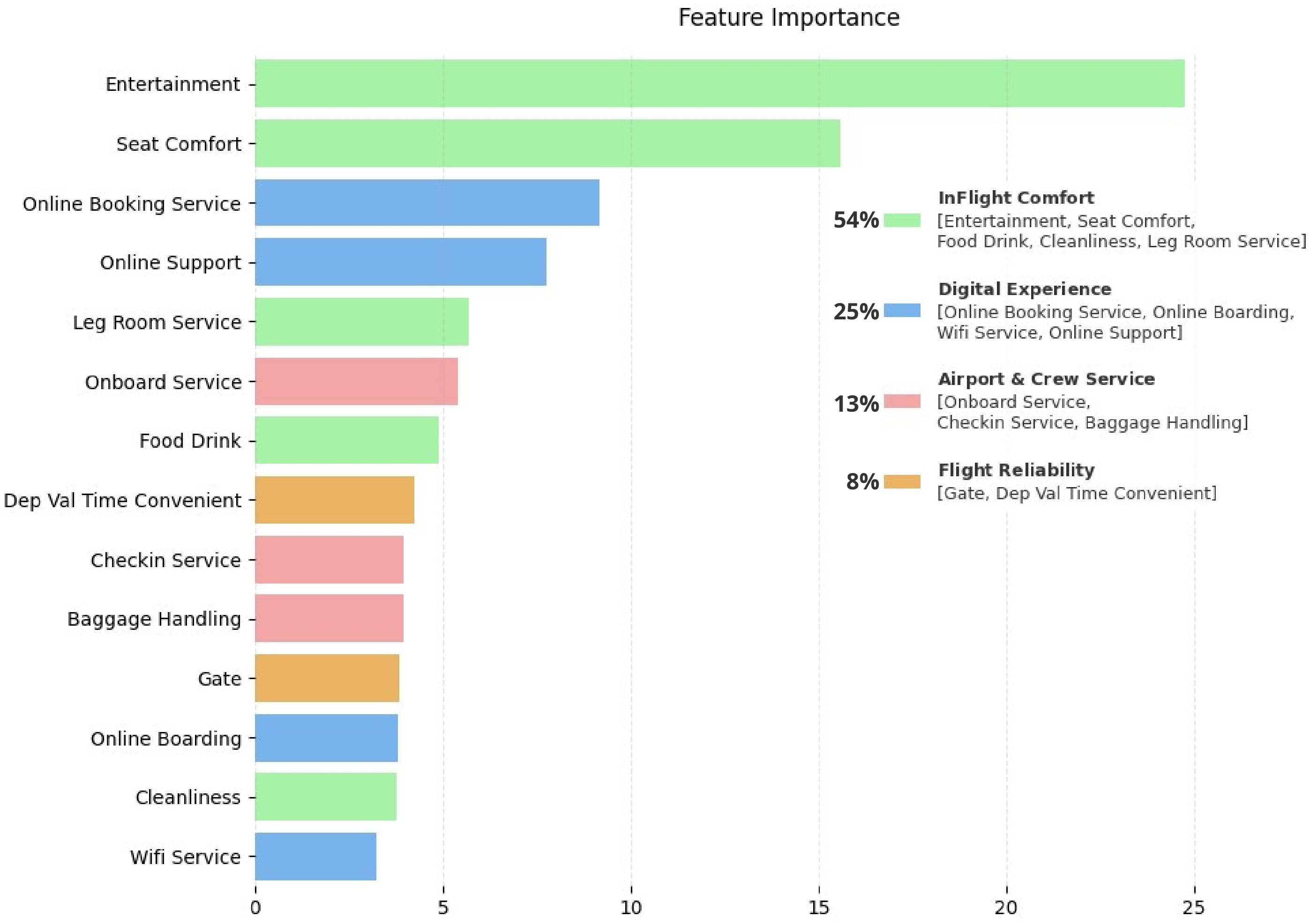
• Personalized offers, retention discounts or miles.

Business Outcome

Strategic & Financial ROI:

• Capital expenditure is optimally allocated only to high-impact features.

Satisfaction Factors: an ML Perspective



- **Strategic Priority:** Retention of loyal passengers depends heavily on InFlight Comfort and Digital Experience.
- **Key Drivers:** The primary satisfaction drivers are Entertainment, Seat Comfort, and Online Services.
- **The main takeaway:** if the entertainment system isn’t working or is boring, passengers will be dissatisfied, even if everything else is good.
- **Hidden Value:** Convenient departure times and optimized gate proximity are latent but important factors.

Hidden Drivers of Loyalty: ML Insight

Statistical Approach

Method

Measures linear dependency only.

Result

Seat Comfort ranks in the middle of the passenger priority list.

Risk

This critical factor may be ignored during budget planning.

ML Modeling

Method

Identifies complex, non-linear dependencies, feature interactions and service synergies.

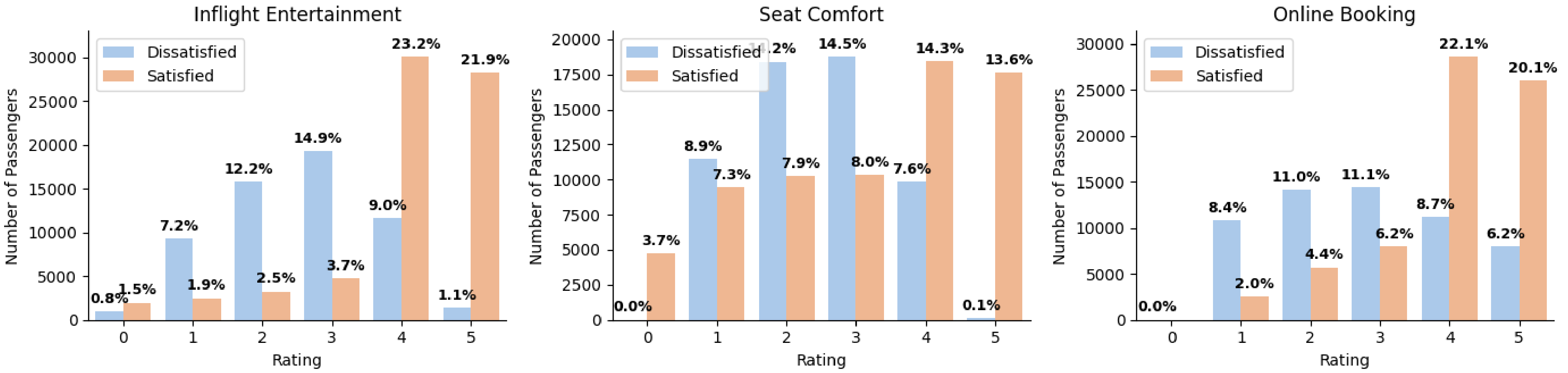
Model Outcome

Seat Comfort ranks among the Top-3 most important features.

Insight

Seat comfort is a baseline expectation for business travelers. Its absence negates the value of all other services.

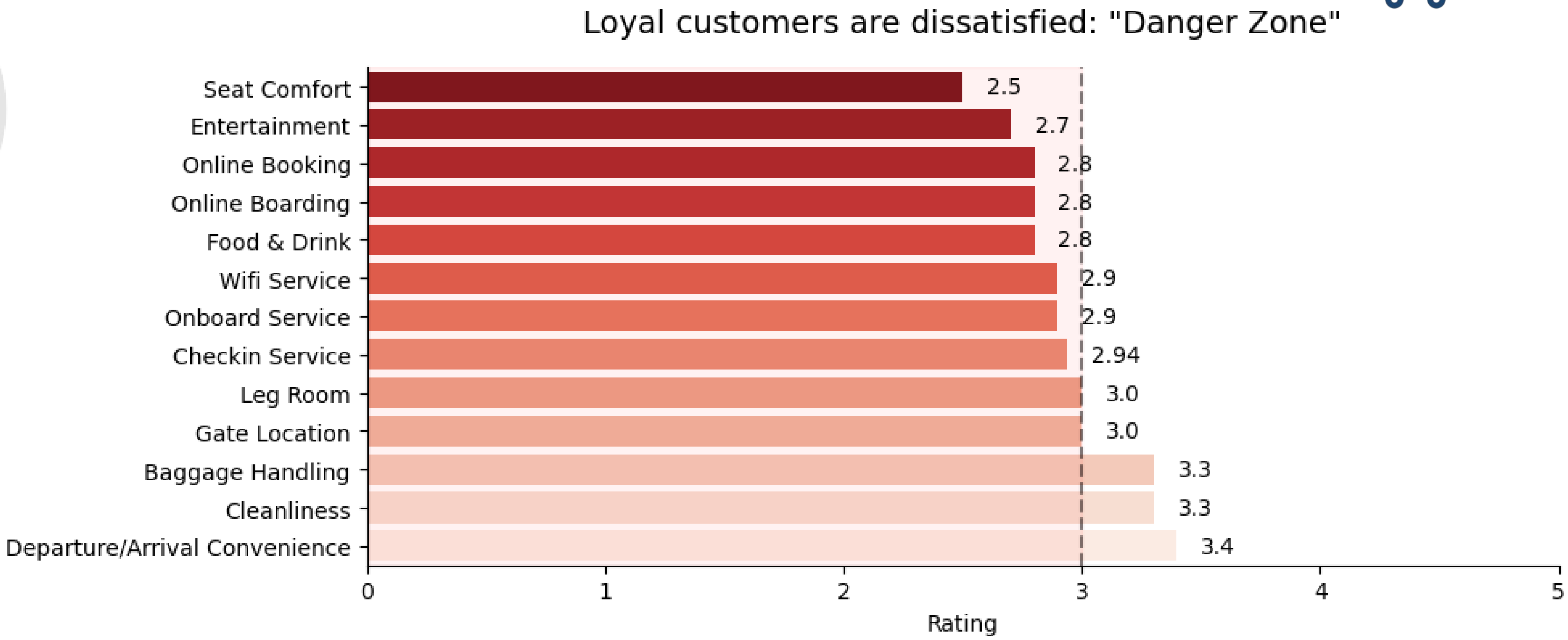
Service Deep Dive: Understanding Loyalty Drivers



- **The 4-Star Threshold:** All tree services, satisfaction scales significantly only when ratings reach 4 or 5 .
- **Entertainment:** Satisfaction jumps from 3.7% to 23.2% at rating levels 3 to 4. Ratings 4–5 drive a 45.1% satisfaction.
- **Seat Comfort:** There is also a jump from dissatisfaction to satisfaction at a rating of 4–5.
- **Online Booking:** Dissatisfaction 11.1% even at a 3-star rating. Passenger loyalty triggers only at levels 4 & 5 - 42.2%.
- **Conclusion:** Passengers perceive a 3-star service almost as negatively as a 1-star. To increase retention, the airline must consistently deliver 4-to-5-star experiences in these critical categories.

Loyal but dissatisfied: Zones of highest risk

69.1%
loyal but dissatisfied



- **High Alert:** Features with scores below 3.0 represent a churn risk zone for loyal passengers.
- **Critical:** The top 3 most important features are currently at the highest risk. The performance target for these metrics is 4.0+.
- **Action:** Upgrade cabin seating and accelerate investments in InFlight Entertainment and Digital Service infrastructure.

Conclusions

Core Customer Profile

- High-Value Segment: Established professionals aged 38–60 flying Business Class form the backbone of airline revenue, representing 82% of the loyal customer base.
- Churn Risk: Despite their overall loyalty, 31% passengers are dissatisfied, creating a hidden risk. Target KPI: customer retention.

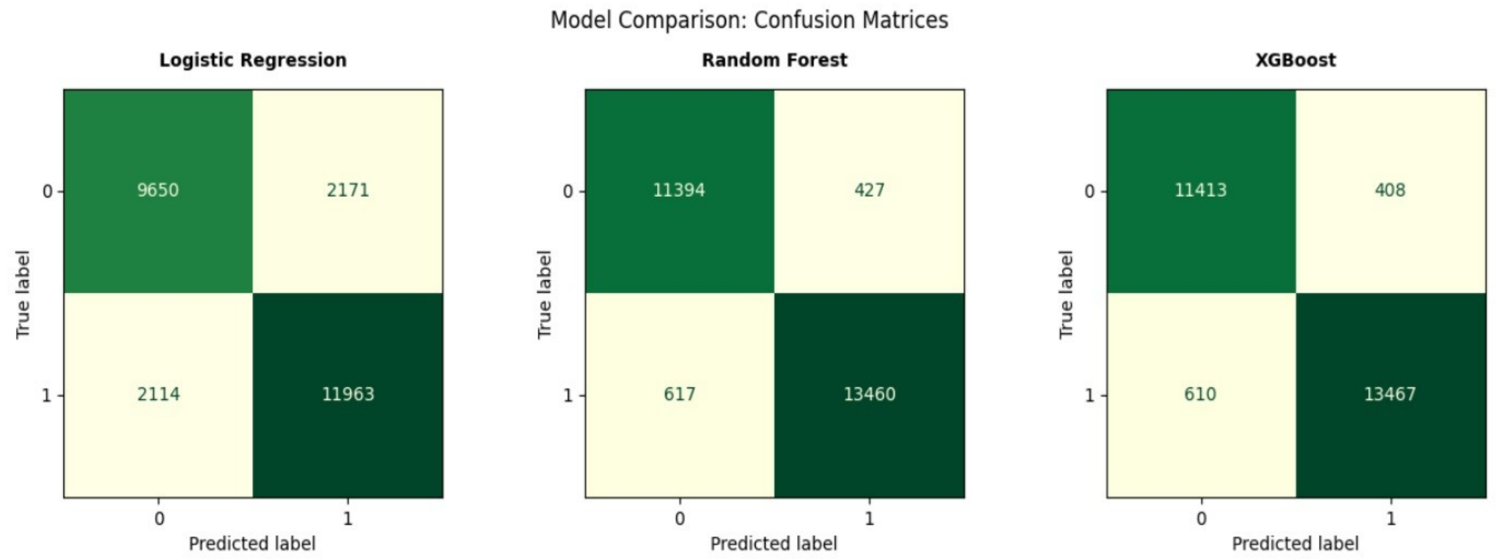
Key Friction Points & Infrastructure Risks

- 4.0+ Benchmark: The main satisfaction drivers are In-Flight Comfort: Entertainment, Seat comfort and Digital Experience: Online Booking. Budget allocation must target these areas, as only a 4.0+ rating triggers true customer satisfaction. Prioritise the updating of digital content and onboard screens.
- Operational Thresholds: Delays exceeding 15 minutes trigger dissatisfaction. After a 2-hour delay, customer satisfaction drops.

Strategic Takeaways (ROI-Driven Retention)

- Baseline Expectation: For this high-value audience, superior comfort and flawless digital services are standard requirements, not premium bonuses.
- Business Case: Acquiring a new customer costs 5 to 7 times more than retaining an existing one. Maintaining high performance in core service features is essential to secure loyal passenger base.

Model Selection: Random Forest vs XGBoost



	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Type
2	XGBoost	96	97	96	96	99	Baseline
1	Random Forest	96	97	96	96	99	Baseline
0	Logistic Regression	83	85	85	85	91	Baseline

- **Collinearity Resilience**: Random Forest provides well-distributed and stable feature importances, preventing highly correlated attributes from artificially neutralizing each other.
- **Robustness to Overfitting**: Independence of decision trees ensures structural stability on skewed customer profiles, mitigating the risks of algorithmic overfitting.
- **Business Interpretability**: Delivers more realistic, transparent, and actionable insight weights for targeted corporate retention campaigns.

By the way, I have deployed this model as a live web application, so we can simulate its business impact right now.